

SURIYAN DHANAPAL

Full-Stack Software Engineer | AI/LLM Integration

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SUMMARY

Full-stack Software Engineer with around 2 years shipping production systems end to end — from React/Next.js frontends to Node.js/FastAPI backends — with deep experience in authentication, real-time infrastructure, and AI/LLM integration. Built and scaled a real-time voice-AI provider layer integrating 25+ STT, TTS, and LLM services behind a unified connector interface, while owning the full frontend-to-backend stack for auth, knowledge ingestion, webhooks, and payments. Comfortable taking ambiguous, infrastructure-heavy problems from design to production on both the client and server side.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frontend: React, Next.js, Redux Toolkit, React Native (Expo), Vite, Tailwind CSS

Backend & System Design: Node.js, Express, FastAPI, REST API design, distributed webhook delivery, idempotency & retry design, OAuth 2.0, JWT auth, role-based access control, gRPC, WebSockets, WebRTC/SIP basics

Data & Infra: MongoDB, PostgreSQL, Cassandra, Redis, pgvector, Docker, GCP, AWS (S3, Bedrock), GitHub Actions (CI/CD)

AI & Integration: LLM integration (OpenAI, Google Vertex AI/Gemini, AWS Bedrock, Groq, etc.), AI agents, LangGraph, embeddings, RAG/retrieval, vector search (Weaviate), STT/TTS pipelines, STS, OCR, browser automation

Tools & Practices: Git, end-to-end feature testing, smoke testing (Claude Code-assisted), Claude Code

PROFESSIONAL EXPERIENCE

Software Engineer — Mindgryd Software Private Limited

Jun 2025 – Aug 2026

AI agent platform · Coimbatore, India

Workflow Automation Platform

Tech: Next.js, TypeScript, Redux Toolkit, Tailwind CSS · FastAPI (Python), OAuth 2.0, Redis, RAG, LangGraph, Weaviate, embedding models, GCP · Node.js/Express, MongoDB, Cassandra, AWS (S3, Bedrock)

- **Build the full-stack workflow platform end to end:** a Next.js/Redux Toolkit frontend for configuring and managing AI agents and automated workflows, backed by Node.js/Express and FastAPI services, including permission controls for how much autonomy an agent has before an action requires human approval.
- **Built the payments layer across the stack:** integrated Razorpay and PayPal (checkout UI through backend), including webhook signature verification and idempotent transaction handling, and designed the core MongoDB schemas powering the payment layer.
- **Built a workspace-specific AI agent recommendation system:** embedded workspace details captured during onboarding, indexed them in Weaviate, and retrieved/ranked relevant agents at search time from a pool of thousands of personalized agents, surfaced through the Next.js frontend.

AI Memory & Context Platform

Tech: FastAPI (Python), PostgreSQL, pgvector, SQLAlchemy, embeddings, RAG · Next.js frontend

- **Built the authentication and onboarding system end to end:** OAuth 2.0 sign-in with Google, GitHub, and Microsoft plus email/password, JWT session handling, role-based access control, and automatic organization provisioning on first sign-in, across both the Next.js frontend and FastAPI backend.
- **Built the knowledge-ingestion subsystem, backend and frontend:** 8 connectors including Google Drive, Dropbox, Notion, and SharePoint, handling per-user OAuth token storage/refresh and content normalization.
- **Built a webhook delivery service:** HMAC-signed payloads, at-least-once delivery via idempotency keys, and exponential-backoff retries, eliminating duplicate-processing incidents for downstream integrators.

Real-time Voice-AI Agents Platform

Tech: Vite + React, FastAPI (Python), WebRTC, SIP, gRPC, PostgreSQL, Redis, FlowCat (in-house open-source real-time media engine)

- **Integrated 25+ STT, TTS, STS, and LLM providers** (OpenAI, Sarvam, Whisper, ElevenLabs, Groq, Gemini, Vertex AI, and others) behind a unified connector interface on FlowCat, our in-house, open-sourced real-time media engine, cutting new-provider integration time from days to hours.
- **Debugged provider-specific failures** affecting live calls (gRPC streaming, tool-call mapping, API-key resolution) and built provider-aware test flows to catch regressions before release.
- **Built a Vite + React dashboard** for real-time monitoring and debugging of live voice-AI calls, replacing raw-log digging with a single view of call status, provider health, and test-flow results — the team's go-to tool for triaging live call issues.

Software Engineer — Hifresh Agro Private Limited

Sep 2024 – Jun 2025

Agro-tech product company · Coimbatore, India (same founding team; moved to Mindgryd when it was spun up)

Tech: React Native (Expo), Redux · Node.js/Express · Zoho Books API, ESC/POS thermal printers, LRU caching

- **Built and owned a full-stack internal delivery-management mobile app:** React Native/Expo frontend used daily by the company's delivery team end to end, extending the existing Node.js/Express backend.
- **Developed APIs for delivery assignment, billing, and invoicing** (Zoho Books integration) and integrated ESC/POS thermal printers for in-app receipt printing.
- **Added response caching and structured logging** to the Express backend's highest-traffic endpoints, eliminating redundant database queries on the app's most-used routes.

CONTRACT / PROJECT WORK

Cooper Compass — Document & Email Intelligence for Logistics (IIT Bombay-incubated startup)

coopercompass.com

Tech: Node.js, Express, TypeScript, OpenAI, Google Vision OCR, Tesseract.js, OAuth 2.0 (Gmail / Microsoft Graph), PostgreSQL, Redis, Docker, GitHub Actions

- **Built a full-stack email-intelligence platform** that cut manual extraction of job/BL numbers from shipping emails from ~2 hours to ~8 minutes per shipment across 20+ pilot users.
- **Built real-time email ingestion** from Gmail and Outlook (OAuth 2.0 with Gmail Pub/Sub and Microsoft Graph push notifications), with Server-Sent Events driving live dashboard updates on the frontend.
- **Implemented the AI document-extraction pipeline:** Google Vision OCR with a Tesseract.js fallback, OpenAI-based classification, and a regex validation layer for job, BL, AWB, and port fields, auto-grouping documents by shipment.

EDUCATION

Bachelor of Engineering — Electronics and Communication Engineering

Dr. N.G.P. Institute of Technology · 2019 – 2023